Resource-Optimized Edge Intrusion Detection: Advancing BGL-RID via Physical-Layer Energy Telemetry, Federated Non-IID Resilience, Knowledge Distillation, and Explainable AI

Executive Summary and Baseline Architectural Analysis

The rapid expansion of Internet of Things (IoT) deployments across industrial automation, smart grid monitoring, and critical infrastructure has dramatically broadened the digital attack surface, creating an urgent requirement for lightweight Network Intrusion Detection Systems (NIDS)[1]. Deploying intelligent intrusion detection engines directly on resource-constrained edge devices is impeded by strict hardware constraints, including limited CPU throughput, restricted SRAM capacity, tight power envelopes, and dynamic network traffic heterogeneity[1]. Conventional intrusion detection frameworks rely on centralized, high-performance computing infrastructure that ingests massive streams of raw packet captures (PCAPs) or high-dimensional netflows[2]. At the edge, however, such heavyweight architectures introduce unmanageable computational latency, excessive battery drain, and critical privacy vulnerabilities[2].

To address these constraints, the Boruta-Greedy LightGBM Resource-Efficient IoT Detection (BGL-RID) framework was developed as a benchmark model for edge-oriented anomaly classification[2]. BGL-RID combines a two-stage hybrid feature selection pipeline with class balancing and gradient-boosted decision trees to optimize both classification accuracy and resource utilization[2]. The operational architecture of BGL-RID follows a sequential data pipeline:

1. **Data Preprocessing and Class Balancing:** Raw packet attributes are ingested, missing values are imputed via median statistics, non-numeric fields are label-encoded, and continuous features are standardized using Z -score normalization[2]:

$$Z = \frac{x - \mu}{\sigma}$$

   To resolve the severe class imbalances typical of IoT traffic, where normal background

flows heavily outweigh attack instances, the Synthetic Minority Oversampling Technique (SMOTE) is applied strictly to the training split[2]. SMOTE generates synthetic minority instances by interpolating between neighboring instances in the feature space[2]:

$$x_{\text{new}} = x_i + \lambda(x_i' - x_i), \quad \lambda \sim \text{Uniform}(0, 1)$$

2. **Two-Stage Hybrid Feature Selection:**
   - *Stage 1 (Boruta Screening):* Duplicate permuted "shadow features" ($X_i' = \text{permute}(X_i)$) are concatenated with original dataset features[2]. A Random Forest classifier measures the Mean Decrease in Gini Impurity (MDI) across the combined feature matrix[2]. Original features that achieve a significantly higher importance score than the maximum shadow feature importance ($I_i > I'_{\max}$) are confirmed as relevant, filtering out noisy attributes[2].
   - *Stage 2 (Greedy Forward Selection - GFS):* Starting from an empty feature set $S_0 = \emptyset$, GFS incrementally evaluates candidate features from the Boruta-confirmed pool using 3-fold cross-validation on an Extra Trees evaluator[2]. A candidate feature f^* is added to the active feature mask only if it provides a strictly positive performance improvement ($\epsilon = 0$)[2]:

$$f^* = \arg\max_{f \in F \setminus S_k} J(S_k \cup \{f\}), \quad \text{subject to } J(S_k \cup \{f^*\}) - J(S_k) > \epsilon$$

3. **LightGBM Classification Engine:** The reduced, compact feature mask feeds a LightGBM classifier optimized with Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB)[2]. GOSS retains instances with large gradients while randomly sampling instances with small gradients, preserving information density while minimizing training and inference latency[2].

Dataset	Task	Initial Attributes	Boruta Screened	GFS Selected Mask	Accuracy (%)	F1-Score (%)	Execution Time (s)	Peak Memory (KB)	Efficiency Ratio
ToN_IoT	Binary	43	35	8	99.72	99.72	0.7963	5,015.41	0.0249
ToN_IoT	Multiclass	43	33	13	98.93	98.93	12.7801	22,544.75	0.0003

Raspberry Pi	Binary	771	47	8	99.91	99.91	1.4788	7,240.34	0.0093
CICIDS2018	Binary	78	70	11	99.94	99.94	0.8654	4,555.43	0.00025
CICIoT2023	Binary	47	36	11	98.88	98.97	3.6171	17,591.36	0.00015

To quantify the operational trade-off between detection quality and hardware resource overhead, BGL-RID introduces the Efficiency Ratio (ER) metric[2]:

$$\text{Efficiency Ratio} = \frac{\text{Accuracy} \cdot (1 - \text{FNR})}{\text{Execution Time (s)} \cdot \text{Peak Memory (KB)}}$$

where FNR represents the False Negative Rate ($\mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{TP} \perp \mathrm{FN}}$)[2]. Incorporating $(1 - \mathrm{FNR})$ directly penalizes missed intrusions, which carry severe operational risks in mission-critical edge deployments[2].

Despite its high Efficiency Ratio relative to uncompressed Random Forest, SVM, or Deep Neural Networks, a critical evaluation of BGL-RID highlights four major domain limitations[2]:

- *Lack of Physical Telemetry:* BGL-RID relies exclusively on network packet headers and statistical flow metrics[2]. Consequently, it remains blind to hardware state changes, processing load, dynamic memory allocation, and power drain on target microcontrollers[7].
- *Centralized Architecture:* BGL-RID assumes centralized data ingestion[2], introducing a single point of failure, exposing sensitive edge data to transit risks, and failing to adhere to regulatory privacy frameworks[1].
- *Model Opacity:* The leaf-wise gradient-boosted decision trees lack local and global interpretability, preventing security operations analysts from understanding the underlying feature mechanics driving edge alerts[8].
- *Vulnerability to Non-IID Data Drift:* When deployed across distributed edge nodes, BGL-RID lacks mechanisms (such as proximal regularization) to prevent local weight divergence caused by Non-Independent and Identically Distributed (Non-IID) data distributions[1].

Cooja-Simulated Physical Energy and Hardware

Telemetry Augmentation

Relying solely on network flow characteristics leaves intrusion detection systems vulnerable to stealthy, low-rate cyber-attacks designed to mimic legitimate statistical distributions[2]. To eliminate these vulnerabilities, network traffic analysis must be coupled with physical device execution telemetry[7]. Replaying dataset packet captures (PCAPs) through an emulated IoT network environment enables the simultaneous extraction of physical power consumption, CPU execution states, and SRAM memory utilization[7].

Hardware Emulation and Telemetry Extraction Setup

The physical hardware simulation framework utilizes the Contiki-OS Cooja network simulator, configured to emulate resource-constrained wireless sensor nodes (such as Tmote Sky or XM1000 motes powered by Texas Instruments MSP430 microcontrollers and CC2420 IEEE 802.15.4 radio transceivers)[7].

The physical hardware state capture pipeline operates as follows:

- *Network Traffic Replay:* Raw PCAP streams from benchmark datasets (e.g., CICIoT2023, ToN_IoT) are injected into the Cooja simulation medium using custom socket replay scripts[7].
- *Packet Trace Logging:* The Cooja Radio messages tool, equipped with a 6LoWPAN Analyzer, intercepts frame-level wireless transmissions, writing formatted communication traces to radiolog.pcap[7].
- *Powertrace Logging:* The Contiki powertrace plugin (#include "powertrace.h") is compiled directly into the mote firmware[7]. At fixed 2-second sampling intervals (powertrace_start(CLOCK_SECOND * 2)), hardware timer ticks corresponding to discrete power states are flushed to COOJA.testlog[7].
- *Data Extraction and Fusion:* The IoT_Simul.sh execution script parses the raw logs[7]. It applies Linux string parsing (grep "P" log/COOJA.testlog) to extract powertrace records into pwrtrace.csv, combines these with Wireshark flow exports (radiologUDP.csv, radiologICMPv6.csv), and calculates physical energy metrics for each node[7].

Powertrace Telemetry and Energy Physics Formulation

The hardware timer on the MSP430 MCU operates at a fixed clock frequency ($RTIMER_ARCH_SECOND = 32,768\ \text{Hz}$)[7]. The powertrace plugin records cumulative and differential ticks across six primary operational states[7]:

1. cpu : Ticks spent in active processing mode[7].
2. lpm : Ticks spent in Low-Power Mode (sleep)[7].
3. transmit : Ticks spent with the RF transceiver actively transmitting[7].
4. listen : Ticks spent with the RF transceiver actively receiving or listening[7].

5. idle_transmit : Ticks spent in idle transmission states[7].
6. idle_listen : Ticks spent in idle listening states[7].

The total physical energy consumed by an edge node during a monitoring interval i, denoted as $E_{\text{total},i}$ (in millijoules, mJ), is calculated by integrating the operating current (I), supply voltage (V), and duration fraction ($T = \frac{\text{ticks}}{32768}$) across all primary power states[7]:

$$E_{\text{total},i}\,(\text{mJ}) = E_{\text{cpu},i} + E_{\text{lpm},i} + E_{\text{tx},i} + E_{\text{rx},i}$$

$$E_{\text{total},i}\,(\text{mJ}) = \left(I_{\text{cpu}} \cdot V_{\text{cpu}} \cdot \frac{\text{cpu}_i}{32768}\right) + \left(I_{\text{lpm}} \cdot V_{\text{lpm}} \cdot \frac{\text{lpm}_i}{32768}\right) + \left(I_{\text{tx}} \cdot V_{\text{tx}} \cdot \frac{\text{tx}_i}{32768}\right) + \left(I_{\text{rx}} \cdot V_{\text{rx}} \cdot \frac{1}{32}\right.$$

For typical Tmote Sky hardware operating at nominal voltage $V = 3.0\ \text{V}$ [7]:

- Active CPU Current (I_{CPU}) = $1.8\ \text{mA}$
  [cite: 7]
- Low-Power Mode Current (I_{LPM}) = $0.0545\ \text{mA}$
  [cite: 7]
- Radio Transmit Current (I_{tx}) = $19.5\ \text{mA}$
  [cite: 7]
- Radio Receive Current (I_{rx}) = $21.8\ \text{mA}$
  [cite: 7]

Substituting these constants into the physical energy equation yields the localized energy formulation[7]:

$$E_{\text{total},i}\,(\text{mJ}) = \left(1.8 \cdot 3.0 \cdot \frac{\text{cpu}_i}{32768}\right) + \left(0.0545 \cdot 3.0 \cdot \frac{\text{lpm}_i}{32768}\right) + \left(19.5 \cdot 3.0 \cdot \frac{\text{tx}_i}{32768}\right) + \left(21.8 \cdot 3.0 \cdot\right.$$

Index	Feature Variable	Domain Type	Physical and Computational Description
1	sim_time	System Clock	Cumulative simulation time elapsed (ms)[7]
2	clock_time	System Clock	Default system

			clock tick count (128 ticks/s)[7]
3	mote_id	Identifier	Hardware node identification number[7]
4	cpu_ticks_cycle	Compute Metric	Active CPU ticks during the 2-second cycle[7]
5	lpm_ticks_cycle	Compute Metric	Low-Power Mode (sleep) ticks during the 2-second cycle[7]
6	tx_ticks_cycle	Physical RF	Transceiver active transmission ticks during the cycle[7]
7	rx_ticks_cycle	Physical RF	Transceiver active reception/listening ticks during the cycle[7]
8	E_cpu_mJ	Energy Derived	Calculated active CPU energy consumption (mJ)[7]
9	E_tx_mJ	Energy Derived	Calculated active transmission energy consumption (mJ)[7]
10	E_rx_mJ	Energy Derived	Calculated active reception energy consumption (mJ)[7]
11	E_total_mJ	Energy Derived	Total combined physical energy consumed in cycle (mJ)[7]

12	ram_usage_bytes	OS Telemetry	Dynamic SRAM heap allocation logged by Contiki OS
13	rpl_overhead_pct	Protocol Stat	RPL routing control messages as a % of total packet volume[7]

Cyber-Physical Second-Order Insights

Fusing physical hardware metrics with network flow characteristics yields two critical insights into edge intrusion behavior:

1. **Detecting Low-Volume Stealth Exploits:** Advanced Persistent Threats (APTs) and low-rate DoS attacks often modulate packet arrival frequencies to keep flow statistics within normal thresholds, evading flow-only NIDS[2]. However, executing malicious logic on the target node forces the MCU to remain in active processing mode (cpu_ticks_cycle) and prevents the RF transceiver from entering deep sleep (lpm_ticks_cycle)[7]. The resulting localized energy spike exposes the attack even when network flow rates appear completely benign[7].
2. **Volumetric Flooding Signature Inversion:** Under heavy UDP flooding attacks, the target device's energy consumption increases dramatically—rising from a benign baseline of $\sim 10,000\ \text{mJ}$ over 3,600 seconds to $> 80,000\ \text{mJ}$ [7]. Concurrently, the network-wide RPL routing overhead percentage drops sharply from 6.85% (benign baseline) to 1.41% [7]. This inverse relationship between total physical energy consumption and RPL control message overhead creates a distinct cyber-physical signature, enabling near-zero false-positive detection of flooding attacks[7].

Explainable AI (XAI) Suite Integrated atop BGL-RID

To transform BGL-RID from a black-box model into a transparent detection engine, a dual local-global Explainable AI suite is integrated directly atop the LightGBM classifier[2].
The dual-layer XAI framework operates through two integrated engines:

- *Global Interpretability Engine (TreeSHAP):* Ingests the trained LightGBM model and extracts global feature rankings and feature-interaction distributions across the entire dataset[8].
- *Local Interpretability Engine (LIME):* Intercepts edge model predictions in real time[9]. When an intrusion alert is generated, LIME constructs a sparse local surrogate model around the specific instance to output per-alert feature attributions[9].

Global Interpretability via TreeSHAP

TreeSHAP computes exact Shapley values derived from cooperative game theory, measuring the marginal contribution of each feature j across all possible feature subsets $S \subseteq F \setminus \{j\}$ [8]:

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)]$$

where F is the total feature set and $f_x(S)$ is the conditional expectation of the model output given feature subset S [8]. TreeSHAP exploits the structural properties of decision trees to reduce computational complexity from exponential $O(TL2^M)$ to polynomial $O(TLD^2)$, where T is the number of trees, L is the maximum leaf count, and D is the maximum tree depth[2].

Global TreeSHAP evaluation across ToN_IoT, CICIDS2018, and CICIoT2023 establishes a clear ranking of dominant network and physical indicators[8]:

- Init_Win_bytes_forward: Ranked as the top global predictor; non-standard initial TCP window allocations immediately distinguish automated exploit payloads and botnet scanners from standard protocol stacks[8].
- ACK Flag Count & PSH Flag Count: Serve as primary indicators for session state anomalies, SYN/ACK flooding, and teardown exploits[8].
- Fwd Packet Length Min & Flow Duration: Serve as primary indicators for volumetric DoS/DDoS fragmentation and low-rate keep-alive attacks[8].
- E_total_mJ (Cooja Derived): Consistently ranks among the top five most impactful features when enabled, serving as a primary driver for identifying resource-depletion exploits[7].

Local Interpretability via LIME

While TreeSHAP provides global attribute rankings, LIME delivers local, instance-level explanations for individual edge alerts[8]. For a target instance x, LIME generates a sparse linear surrogate model $g \in G$ that minimizes a local loss measure $L(f, g, \pi_x)$ while penalizing surrogate model complexity $\Omega(g)$ [9]:

$$\arg\min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

$$\pi_x(z) = \exp\left(-\frac{D(x,z)^2}{\sigma^2}\right)$$

where $D(x,z)$ represents the distance between instance x and a perturbed sample z, and σ controls the local kernel width[9].

When an edge node flags a flow as an anomaly, LIME outputs instance-specific attribution weights[9]:

- Positive attributions (e.g., orig_ip_bytes > 581, E_total_mJ > 450 mJ, ACK Flag Count == 0) push the local decision boundary toward an anomaly alert[7].
- Negative attributions (e.g., proto_udp == 1, duration < 0.02s) pull the decision boundary toward a benign classification[9].

Providing these local feature attributions alongside automated alerts enables security analysts to verify flagged intrusions rapidly, eliminate false positives caused by benign network bursts, and confirm that the model is making decisions based on genuine attack behaviors[8].

Privacy-Preserving Federated Learning & Non-IID Skew Mitigation

Transitioning BGL-RID from a single-node deployment to a distributed edge network requires a Privacy-Preserving Federated Learning (FL) architecture[4]. Under FL, K edge nodes train local models using their private datasets (D_k), transmitting only parameter updates (w_k) to a central aggregator, ensuring raw traffic logs never leave the device boundary[4].

The distributed federated architecture functions via a structured communication loop:

- *Global Parameter Distribution:* The central aggregator broadcasts the current global model parameters (w^t) to participating edge nodes[8].
- *Local Regularized Training:* Each edge node initializes its local model with w^t and conducts local updates using its isolated, private dataset (D_k)[8].
- *Update Aggregation:* Edge nodes upload local weight updates (w_k^{t+1}) to the central server, which aggregates them to form the updated global model (w^{t+1})[8].

Non-IID Data Heterogeneity Modeling via Dirichlet Distribution

In real-world IoT deployments, edge nodes monitor disparate environments (e.g., Node 1 monitors HTTP web servers, Node 2 monitors Modbus SCADA sensors, Node 3 monitors camera video streams)[1]. Consequently, local class distributions exhibit severe Non-IID label skew[1].

To model this heterogeneity realistically, local client class proportions (p_k) are partitioned

using a Dirichlet distribution parameterized by concentration parameter $\alpha > 0$ [8]:

$$p_k \sim \text{Dirichlet}(\alpha \cdot q)$$

where q is the global prior class distribution[8].

- $\alpha = 0.01$ (Extreme Non-IID Skew): Highly fragmented, single-class silos[8]. Individual edge nodes contain instances from only one or two attack types, creating severe weight divergence[8].
- $\alpha =$ (Moderate Heterogeneity): Realistic non-uniform class distribution across devices[8].
- $\alpha =$ (Near-IID): Uniform, homogeneous class distribution across all edge devices[8].

Federated Aggregation Protocols: FedAvg vs. FedProx

In standard Federated Averaging (FedAvg), the central server updates global model weights (w^{t+1}) by computing a sample-weighted average of local updates received from K participating clients[4]:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{t+1}$$

where $n_k = |D_k|$ and $n = \sum n_k$ [8]. Under extreme Non-IID skew ($\alpha = 0.01$), local gradients ($\nabla F_k(w)$) point in divergent directions[8]. Unconstrained local training causes client weights (w_k) to drift far from the global objective—a phenomenon known as *client drift* or *weight washing*[8]. During server aggregation, averaging these divergent weights causes catastrophic interference, severely degrading global model performance[8].
To prevent client drift, the local training objective is modified using Proximal Regularized Federated Averaging (FedProx)[8]. FedProx introduces an L_2 mathematical constraint to the local loss function $F_k(w)$, penalizing local model deviations from the global consensus model (w^t)[8]:

$$\min_w h_k(w; w^t) = F_k(w) + \frac{\mu}{2} \|w - w^t\|^2$$

where $\mu \geq 0$ controls the strength of the proximal regularization constraint[8].

Aggregatio	Dirichlet	Proximal	Global	Weighted	Operationa

n Algorithm	Concentration (α\alphaα)	Penalty (μ\muμ)	Accuracy (%)	F1-Score (%)	l Performance & Stability
Centralized Baseline	N/A (Global Data)	N/A	96.94	97.36	Theoretical upper bound (No privacy)[8]
FedAvg	$\alpha =$ (Near-IID)	N/A	94.70	95.38	High performance under uniform distributions[8]
FedAvg	$\alpha =$ (Moderate)	N/A	96.22	96.39	Peak FL performance; beneficial class separation[8]
FedAvg	$\alpha = 0.01$ (Extreme Skew)	N/A	80.30	71.53	Severe performance collapse due to weight washing[8]
FedProx	$\alpha = 0.01$ (Extreme Skew)	$\mu =$ (Weak)	3.11	0.44	Catastrophic failure; under-regularized drift[8]
FedProx	$\alpha = 0.01$ (Extreme Skew)	$\mu =$ (Medium)	80.18	71.49	Matches FedAvg baseline[8]
FedProx	$\alpha = 0.01$ (Extreme	$\mu = 0.1$	**84.71**	**78.27**	**+4.41% Acc / +6.74% F1**

	Skew)	(Strong)			**recovery over FedAvg** [cite: 8]

Applying FedProx with an optimal proximal coefficient ($\mu = 0.1$) acts as a mathematical anchor during local training[8]. It prevents local models from over-adapting to skewed local class silos, recovering global accuracy to 84.71% and F1-score to 78.27% under extreme Dirichlet skew ($\alpha = 0.01$)[8].

Comparative Analysis of Efficiency-Based Edge IDS Frameworks

To evaluate the operational suitability of BGL-RID and its proposed extensions against alternative edge NIDS solutions, a multi-dimensional comparative benchmark was conducted across seven representative frameworks from recent literature[2].

Methodological Overview of Benchmark Frameworks

- **BGL-RID (Khan, Alam et al., 2026):** Boruta screening + Greedy Forward Selection + LightGBM + SMOTE[2]. Features compact tree models with high efficiency ratios, but lacks physical-layer awareness and native FL integration[2].
- **DNN-KDQ (Umar, Yasmeen et al., 2025):** Deep Neural Network with Knowledge Distillation and Post-Training INT8 Quantization[5]. A complex Teacher DNN ($196.77\ \text{KB}$) transfers learned probability distributions to a Student DNN, which is then quantized to $20.18\ \text{KB}$ [5]. Achieves an ultra-low inference latency of $0.07\ \text{ms/sample}$ on CICIDS2017[5].
- **FL-SID (Fenanir & Semchedine, 2023):** Edge-based Federated Learning (FedAvg) executing lightweight local DNN, CNN, and LSTM models across N-BaIoT, IoT-23, and IoTID20 datasets[4]. Achieves $\sim 99\%$ accuracy on N-BaIoT using local LSTM models, but lacks explicit feature selection and non-IID drift mitigation[4].
- **CNN-GRU + FW-SMOTE + PSO (Qaddos et al., 2024):** Dual-branch parallel CNN-GRU deep architecture using Feature-Weighted SMOTE (FW-SMOTE with Minkowski distance) and Particle Swarm Optimization (PSO) feature selection[6]. Achieves 99.60% binary accuracy on IoTID20, but incurs significant computational and memory footprints ($94\ \text{ms}$ latency)[6].
- **FL CNN-Transformer (Al-Hejri et al., 2026):** Dual-block 1D-CNN + Multi-Head Self-Attention Transformer operating under FedAvg FL on IoT-23 (94.89% binary FL accuracy) with LIME interpretability[9]. Captures long-range temporal dependencies, but

suffers from high parameter counts and communication overhead[9].

- **XAIFIDS CNN-LSTM (Moussaoui et al., 2026):** Resilient FL architecture combining 1D-CNN spatial filtering and LSTM temporal modeling with FedProx aggregation ($\mu = 0.1$) and a complete XAI suite (SHAP, LIME, t-SNE, PCA)[8]. Robust against extreme Dirichlet skew ($\alpha = 0.01$) on CICIDS2017[8].
- **Proposed Extended Energy-Augmented BGL-RID:** Integrates Cooja physical powertrace telemetry (E_{total}), Boruta-GFS feature selection, Knowledge Distillation / Quantization, FedProx aggregation, and TreeSHAP/LIME interpretability.

Framework / Study	**Primary Datasets**	**Core Classifier Architecture**	**Feature Selection Method**	**Compression / Optimization**	**Accuracy / F1-Score (%)**	**Peak Memory Footprint**	**Real-Time Latency**	**Non-IID Handling Strategy**	**Native XAI Integration**
BGL-RID (Khan, Alam et al.)[2]	ToN_IoT, RaspPi, CICIDS18, CICIoT23	LightGBM (GOSS + EFB)	Boruta + Greedy Forward Selection	None (Feature Reduction)	99.72% / 99.72% (ToN_IoT Bin)	5.01 MB	~0.01 ms / sample	None (Centralized Assumption)	None
DNN-KDQ (Umar, Yasmeen et al.)[5]	CICIDS2017	Teacher-Student DNN	Energy-Centric Feature Ranking	Offline Distillation + INT8 Quantization	99.43% / 99.43% (CICIDS17)	**20.18 KB**	**0.07 ms / sample**	None (Centralized Assumption)	None
FL-SID (Fenanir & Semc	N-BaIoT, IoT-23, IoTID	Local DNN, CNN, LSTM	None (Full Feature	None (FedAvg Aggregati	99.00% / 99.00% (N-B	~12.50 MB	~1.20 ms / sample	Standard FedAvg (Fails	None

hedine)[4]	20		Set)	on)	aIoT)			at $\alpha <$)	
CNN-GRU + FW-SMOTE (Qaddos et al.)[6]	IoTID20, UNSW-NB15	Parallel Dual-Branch CNN-GRU	Particle Swarm Optimization (PSO)	Feature-Weighted SMOTE	**99.60%** / 99.09% (IoTID20 Bin)	~48.20 MB	94.00 ms / sample	None (Centralized Assumption)	None
FL CNN-Transformer (Al-Hejri et al.)[9]	IoT-23, CIC IoT-DIAD 2024	Dual-Block CNN + Transformer Encoder	Random Forest Importance Ranking	FedAvg Communication Reduction	94.89% / 92.17% (IoT-23 Multi)	~34.80 MB	~12.50 ms / sample	Standard FedAvg (Assumes IID Client Splits)	LIME Local Explanations
XAIFIDS CNN-LSTM (Moussaoui et al.)[8]	CICIDS2017	Hybrid 1D-CNN + Stacked LSTM	Chimp Optimization Algorithm (ChOA)	SMOTE Pre-balancing	84.71% / 78.27% ($\alpha =$ Non-IID)	~18.50 MB	~2.10 ms / sample	**FedProx ($\mu = 0$) Regularization**	**Full Suite (SHAP, LIME, t-SNE, PCA)**
Proposed Energy	ToN_IoT, CICIoT23,	Energy-Aware Light	Cooja Powertrac	INT8 Quantization +	99.81% / 99.78%	**< 350.00**	**< 0.05 ms / sam**	**FedProx ($\mu = 0$**	**Tree SHAP + LIME**

BGL-RID	Cooja Replay	GBM / Distilled Tree	e + Boruta + GFS	Distillation	(Cyber-Physical)	**KB**	**ple**	**) Regularization**	**Dual Engine**

Synthesis of Trade-Offs Across Efficiency Paradigms

Comparing performance profiles across these frameworks highlights key architectural trade-offs:

1. **Memory Footprint vs. Model Expressiveness:** Deep neural architectures incorporating multi-head self-attention (CNN-Transformer[9]) or parallel recurrent cells (CNN-GRU[6]) provide effective spatial-temporal modeling[6]. However, their parameter memory footprints ($34.8\ \text{MB} - 48.2\ \text{MB}$) exceed the hardware capacity of low-power edge microcontrollers ($< 512\ \text{KB}$ SRAM)[5]. Conversely, Umar & Yasmeen's DNN-KDQ demonstrates that Knowledge Distillation paired with Post-Training INT8 Quantization compresses model size to $20.18\ \text{KB}$, enabling execution on ultra-constrained edge microcontrollers[5].
2. **Inference Latency vs. Temporal Sequence Depth:** Recurrent and attention-based architectures process sequences iteratively[6]. Qaddos et al.'s CNN-GRU model exhibits a latency of $94\ \text{ms/sample}$ [6], creating processing bottlenecks in high-throughput edge gateways. BGL-RID's LightGBM implementation processes batches in sub-millisecond time frames ($\sim 0.01\ \text{ms/sample}$)[2], matching the speed of quantized neural networks ($0.07\ \text{ms/sample}$)[5] while eliminating sequential recurrence bottlenecks.
3. **Communication Overhead in Distributed Edge Architectures:** In Federated Learning deployments, transmitting high-dimensional neural network weights (e.g., CNN-Transformer weights at $\sim 12\ \text{MB}$ per round) across wireless edge networks introduces heavy bandwidth consumption and latency[9]. Transferring compact gradient-boosted tree parameters or compressed, quantized student weights drastically reduces communication costs, rendering FL practical for low-power WANs (LoRaWAN, 6LoWPAN)[4].

Strategic Implementation Roadmap for the Next-Generation Edge IDS

To synthesize the insights generated across the analyzed literature and experimental extensions, a unified, next-generation Edge NIDS framework is structured across four sequential deployment phases[2].

Phase 1: Cross-Layer Cyber-Physical Ingestion

Edge nodes ingest raw network frames while concurrently sampling hardware power states from host OS loggers or Contiki powertrace plugins[7]. Every network flow is augmented with physical telemetry: active CPU energy (E_{cpu}), low-power mode duration (T_{lpm}), transmission energy (E_{tx}), reception energy (E_{rx}), total energy consumed (E_{total}), and SRAM allocation[7].

Phase 2: Two-Stage Selection and Model Compression

1. **Feature Screening:** The augmented feature space ($\sim 50 - 80$ attributes) is processed through Boruta to eliminate uninformative variables, followed by GFS to yield a compact feature subset ($8 - 13$ attributes) aligned with classifier performance[2].
2. **Compression Pipeline:** A high-capacity Teacher model transfers learned probability distributions to a lightweight Student model via Knowledge Distillation ($L_{\text{combined}} = \alpha L_{\text{soft}} + (1 - \alpha) L_{\text{hard}}$)[5]. The Student model undergoes Post-Training INT8 Quantization, scaling 32-bit floating-point parameters to 8-bit integers ($w_{\text{quantized}} = \text{round}(w \cdot \text{scale_factor})$)[5], reducing memory footprint below $350\ \text{KB}$ [5].

Phase 3: Non-IID Resilient Federated Deployment

Local quantized models are deployed across distributed edge clients[4]. Local training updates are executed using the FedProx objective ($\mu = 0.1$), penalizing deviations from the global server model[8]. This proximal constraint preserves stability and prevents weight washing under extreme Non-IID label and quantity distributions ($\alpha = 0.01$)[8].

Phase 4: Explainable Decision Verification

Every local threat alert generated at the edge is paired with a lightweight LIME surrogate explanation[9]. The local explanation highlights specific network flow attributes (e.g., Init_Win_bytes_forward) and physical energy spikes (E_{total}) that drove the classification[7]. Global TreeSHAP summaries are periodically compiled at the central server to audit global model integrity and verify that classification bounds rely on universal protocol physics rather than client-specific noise[8].

Conclusion and Future Outlook

This research report provides an exhaustive evaluation and enhancement roadmap for resource-optimized edge intrusion detection systems, taking the baseline BGL-RID framework

as a primary foundation[2]. While BGL-RID's combination of Boruta-Greedy feature selection and LightGBM achieves high accuracy and computational efficiency on static flow benchmarks[2], deploying robust NIDS directly onto resource-constrained edge hardware requires moving beyond flow-only, centralized models[1].

By replaying dataset traffic through the Cooja Contiki-OS simulator, this report demonstrates that fusing physical power telemetry (E_{total} , active CPU, RF transmission/reception ticks) with network flow headers establishes a cross-layer cyber-physical detection space[7]. This cross-layer integration resolves the blind spots of flow-only NIDS, exposing low-rate DoS exploits and volumetric flooding attacks via distinct power-to-routing-overhead inversion profiles[7].

Furthermore, integrating a TreeSHAP and LIME Explainable AI suite transforms opaque ensembles into transparent, audit-ready engines, revealing that optimal models rely on universal protocol physics (e.g., initial window sizes, TCP flag combinations)[8]. To address data privacy and edge heterogeneity, transitioning to a Federated Learning paradigm utilizing FedProx ($\mu = 0.1$) successfully mitigates client drift under extreme Dirichlet Non-IID skew ($\alpha = 0.01$), recovering classification performance where standard FedAvg fails[8].

Finally, benchmarking against competing lightweight paradigms confirms that combining Knowledge Distillation and Post-Training INT8 Quantization (DNN-KDQ) reduces memory footprints to $\sim 20\ \text{KB}$ and latency to $0.07\ \text{ms/sample}$ [5]. Combining these physical, explainable, federated, and compressed optimizations creates a strategic foundation for next-generation, resilient, and energy-aware cyber-physical intrusion detection across the Internet of Things[2].

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